6(a)	• greater bandwidth • less noise • less attenuation **or** fewer repeaters • less crosslinking **or** greater security *Any three points, 1 mark each*	**B3**
6(b)(i)	ratio / dB = 10 lg(P_1 / P_2)	**C1**
	21 = 10 lg [(6.3×10^{-17}) / P] $P = 5.0 \times 10^{-19}$ W	**A1**
6(b)(ii)	attenuation per unit length = (1 / 4.5×10^{3}) $\times$ 10 lg [(9.8×10^{-3}) / (6.3×10^{-17})]	**C1**
	= 0.032 dB km^{-1}	**A1**